

CSCI212 Lab — Quiz #2

Full Name:

EMPLID:

Date:

Score:

1. Write a static function titled *computeMeanCoinFlip()* under the following conditions:
 - Takes a single integer parameter *n*.
 - Initializes an **integer array** of size *n*, denoted *coinFlips*.
 - Populates *coinFlips* with *n* randomly generated 1's or 0's (such that 1 and 0 are heads and tails respectively).
 - Compute and return the mean of *coinFlips*
2. Write a static function titled *totalCharacterCount()* under the following conditions:
 - Asks the user to input **10 words** via **System.out.println()**.
 - Reads input via **Scanner** object.
 - Stores user input into an appropriate array.
 - Computes the total number of characters in each word from the array, storing the total count into a variable denoted *characterCount*.
 - Returns *characterCount*.
3. Suppose you have a 2-D array (a matrix) of integers with dimensions **3x3** (rows x cols), denoted as:

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

In linear algebra, the **trace** of a matrix (e.g. *A*) denoted $tr(A)$, is the sum over the diagonal entries of the matrix. i.e. $tr(A) = a_{11} + a_{22} + a_{33}$. Write a static function titled *trace()* that returns the trace of an **nxn** integer array denoted *A*.